

why $KL(P||Q) \geq 0$?

to show $KL(P||Q) \geq 0$.

let $f(x) = -\ln x$, since $f''(x) = \frac{1}{x^2} > 0$,
 $f(x)$ is strictly convex function.

by Jensen's inequality about convex function,

$$f\left(\sum_{i=1}^N \lambda_i x_i\right) \leq \sum_{i=1}^N \lambda_i f(x_i) \quad \text{where } \lambda_i \geq 0, \sum_i \lambda_i = 1.$$

to interpret inequality as expectation, we consider properties of probability density function.

$$\sum_x p(x) = 1 \quad \text{correspond to} \quad \sum_i \lambda_i = 1$$

$$p(x) \geq 0 \quad \text{correspond to} \quad \lambda_i \geq 0.$$

then expanding sigma to integral gives

$$f\left(\int x p(x) dx\right) \leq E\left[\int p(x) f(x) dx\right]$$

$$\Leftrightarrow f[E[X]] \leq E[f(X)] \quad \text{where } f \text{ is convex function}$$

$f(x) = -\ln x$ gives

$$-\ln(E[x]) \leq E[-\ln x] \quad (\because -\ln x \text{ is strictly convex})$$

now substituting $\frac{q(x)}{p(x)}$ gives

$$-\ln E\left[\frac{q(x)}{p(x)}\right] \leq E\left[-\ln \frac{q(x)}{p(x)}\right] \quad (\text{where } \frac{q(x)}{p(x)} > 0)$$

$$\Rightarrow -\ln \int p(x) \cdot \frac{q(x)}{p(x)} dx \leq \underbrace{\int p(x) \left(-\ln \frac{q(x)}{p(x)}\right) dx}_{KL(p||q)}$$

$$\Rightarrow -\ln \int q(x) dx = -\ln(1) = 0 \leq KL(p||q)$$

$$\therefore KL(p||q) \geq 0$$

if $p(x) = q(x)$ for all x ,

$$KL(p||q) = \int p(x) \ln \frac{p(x)}{p(x)} dx = 0$$

if $p(x) \neq q(x)$ then $KL(p||q) > 0$ because $\ln \frac{p(x)}{q(x)} > 0$ for some x .

if $p(x) < q(x)$ then $\ln \frac{p(x)}{q(x)} < 0$ for some x and $\ln \frac{p(x)}{q(x)} > 0$ for some other x .

if $p(x) > q(x)$ then $\ln \frac{p(x)}{q(x)} > 0$ for some x and $\ln \frac{p(x)}{q(x)} < 0$ for some other x .

if $p(x) = q(x)$ then $\ln \frac{p(x)}{q(x)} = 0$ for all x .

if $p(x) \neq q(x)$ then $\ln \frac{p(x)}{q(x)} \neq 0$ for some x .

if $p(x) < q(x)$ then $\ln \frac{p(x)}{q(x)} < 0$ for some x .

if $p(x) > q(x)$ then $\ln \frac{p(x)}{q(x)} > 0$ for some x .

to show $H(p||q) = 0 \iff p(x) = q(x)$ for all x

$$p(x) = q(x) \Rightarrow H(p||q) = \int p(x) \ln \frac{p(x)}{q(x)} dx$$

$$= \int p(x) \ln(1) dx = \int 0 dx = 0.$$

① to show $H(p||q)$ is convex with respect to q .
 let $\underline{q} = \lambda \underline{q}_1 + (1-\lambda) \underline{q}_2$: $\lambda \in [0,1]$

Since $-\ln$ is convex function,

$$-\lambda \ln(\lambda \underline{q}_1 + (1-\lambda) \underline{q}_2) \leq -\lambda \ln \underline{q}_1 - (1-\lambda) \ln \frac{\underline{q}_2}{p}$$

add $\lambda \ln p$ both side.

$$-\ln(\lambda \underline{q}_1 + (1-\lambda) \underline{q}_2) + \lambda \ln p \leq -\lambda \ln \underline{q}_1 - (1-\lambda) \ln \frac{\underline{q}_2}{p} + \lambda \ln p$$

$$\Rightarrow -\ln\left(\frac{\lambda \underline{q}_1}{p} + (1-\lambda) \frac{\underline{q}_2}{p}\right) \leq -\lambda \ln \frac{\underline{q}_1}{p} - (1-\lambda) \ln \frac{\underline{q}_2}{p}$$

multiply by $p(x) \geq 0$ and take integral

$$\Rightarrow \int p \ln\left(\frac{\lambda \underline{q}_1}{p} + (1-\lambda) \frac{\underline{q}_2}{p}\right) dx \leq \int -\lambda p \ln \frac{\underline{q}_1}{p} - (1-\lambda) p \ln \frac{\underline{q}_2}{p} dx$$

$$\Rightarrow H(p||\lambda \underline{q}_1 + (1-\lambda) \underline{q}_2) \leq \lambda H(p||\underline{q}_1) + (1-\lambda) H(p||\underline{q}_2)$$

since λ is arbitrary, $H(p||q)$ is convex with respect to $\underline{q}(x)$.

② to show $KL(p||q)$ has only one critical point with respect to $q(x)$.

since $q(x)$ is under constraint of pdf $\int q(x) dx = 1$, we should use Lagrangian.

($KL(p||q)$'s critical point with respect to $q(x)$)

$\Leftrightarrow$ (critical point of $L(q, \lambda)$ under $\int q(x) dx = 1$)

$$L(q, \lambda) = \int p(x) \ln \frac{p(x)}{q(x)} dx - \lambda \left(\int q(x) dx - 1 \right)$$

to find $\frac{\partial}{\partial q(x)}(L(q, \lambda))$, follow those steps.

1. $q \rightarrow q + \varepsilon \nu$ for any $\nu(x)$

$$\Rightarrow \int p \ln \frac{p}{q + \varepsilon \nu} dx - \lambda \left(\int q + \varepsilon \nu dx - 1 \right)$$

2. partially differentiate with respect to ε

$$\Rightarrow \frac{\partial}{\partial \varepsilon} \left(\int p \ln \frac{p}{q + \varepsilon \nu} dx - \lambda \left(\int q + \varepsilon \nu dx - 1 \right) \right)$$

$$\Rightarrow - \int p \cdot \frac{\nu}{q + \varepsilon \nu} dx - \lambda \left(\int \nu dx \right) = - \int \left(\frac{p}{q + \varepsilon \nu} + \lambda \right) \nu dx$$

3. $\varepsilon \rightarrow 0$.

$$\lim_{\varepsilon \rightarrow 0} - \int \left(\frac{p}{q + \varepsilon \nu} + \lambda \right) \nu dx = - \int \left(\frac{p}{q} + \lambda \right) \nu dx$$

4. by fundamental lemma of variation of calculus,

$$-\int \left(\frac{p}{q} + \lambda \right) N dx = 0 \quad \text{for any } N(x)$$

$$\Rightarrow \frac{p}{q} + \lambda = 0.$$

$$\text{since } \int p dx = \int q dx = 1,$$

$$\frac{p(x)}{q(x)} + \lambda = 0 \Rightarrow p(x) = -\lambda q(x)$$

$$\Rightarrow \int p(x) dx = \int -\lambda q(x) dx$$

$$\Rightarrow 1 = -\lambda,$$

since $\lambda = -1$ is uniquely exist where $\frac{\delta}{\delta q} (L(x, \lambda)) = 0$

and there $p = q$.

$KL(p||q)$ has unique critical point

with respect to q where $q = p$.

then we integrates three result we proved,

- 1st, $KL(p||q) \geq 0$ by Jensen's inequality
- 2nd, $KL(p||q)$ is convex with respect to q ,
- 3rd, $KL(p||q)$ has unique critical points with respect to q on $p=q$ and there $KL(p||q)=0$.

We conclude $KL(p||q)$ has absolute minimum on $p=q$ and there $KL(p||q)=0$.

so without proving $p=q \Rightarrow KL(p||q)=0$,

$$KL(p||q)=0 \Leftrightarrow p=q,$$

++ $KL \geq 0$ is deduced only with KL 's convexity and unique critical point but without Jensen's inequality. However, I could plan to show convexity and uniqueness after I proved non negativity.

it was what suggested this part.